

Time Series — Solutions

Practice Exam — Week 5: Trends, Seasonality & SARIMA

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Problem 1 (5 points)

(a) With $\mu_t = a + bt$, linearity of $\nabla = I - B$ gives

$$\nabla X_t = X_t - X_{t-1} = (a + bt + Y_t) - (a + b(t-1) + Y_{t-1}) = b + (Y_t - Y_{t-1}) = b + \nabla Y_t.$$

So differencing collapses the linear trend $a + bt$ to the constant b . Now $\nabla Y_t = Y_t - Y_{t-1}$ is a *fixed finite linear combination* of a stationary process, hence itself stationary (its mean is 0 and its covariance depends on the lag only). The original $\{X_t\}$ is not stationary because $\mathbb{E}[X_t] = a + bt$ depends on t .

(b) With $\mu_t = a + bt + s_t$ and $s_{t+4} = s_t$, apply the seasonal difference first. Using $s_t = s_{t-4}$,

$$\nabla_{(4)} X_t = X_t - X_{t-4} = (a + bt + s_t + Y_t) - (a + b(t-4) + s_{t-4} + Y_{t-4}) = 4b + (Y_t - Y_{t-4}).$$

The period-4 season has been annihilated and the linear trend has collapsed to the constant $4b$. Now apply $\nabla = I - B$, which kills the constant:

$$\nabla \nabla_{(4)} X_t = (4b + Y_t - Y_{t-4}) - (4b + Y_{t-1} - Y_{t-5}) = \boxed{Y_t - Y_{t-1} - Y_{t-4} + Y_{t-5}}.$$

This is a fixed linear combination of $\{Y_t\}$, so its mean is $\mathbb{E}[Y_t - Y_{t-1} - Y_{t-4} + Y_{t-5}] = 0$, and by bilinearity its covariance,

$$\begin{aligned} \text{Cov}(\nabla \nabla_{(4)} X_t, \nabla \nabla_{(4)} X_{t-\tau}) &= 4\gamma_\tau - 2\gamma_{\tau-1} - 2\gamma_{\tau+1} - 2\gamma_{\tau-4} - 2\gamma_{\tau+4} \\ &\quad + \gamma_{\tau-3} + \gamma_{\tau+3} + \gamma_{\tau-5} + \gamma_{\tau+5} + \gamma_{\tau+1-4} + \gamma_{\tau-1+4}, \end{aligned}$$

depends on the lag τ only (no t). Hence $\nabla \nabla_{(4)} X_t$ is weakly stationary; all t -dependence has been removed. (One need not simplify the covariance fully; the key point is that it is a function of τ alone.)

(c) Treat $\mu_t = t^2$ as a deterministic sequence.

Ordinary second difference $\nabla^2 = (I - B)^2$:

$$\nabla^2 t^2 = t^2 - 2(t-1)^2 + (t-2)^2 = t^2 - 2(t^2 - 2t + 1) + (t^2 - 4t + 4) = 2,$$

a constant. (And $\nabla^3 t^2 = \nabla(2) = 0$.) So ∇^q annihilates a polynomial trend of degree $q-1$: here ∇^2 reduces the degree-2 polynomial t^2 to a degree-0 constant, and one further ∇ kills it.

Seasonal difference of period 2, $\nabla_{(2)} = I - B^2$:

$$\nabla_{(2)} t^2 = t^2 - (t-2)^2 = t^2 - (t^2 - 4t + 4) = 4t - 4,$$

which is still linear (not annihilated). The seasonal operator $\nabla_{(s)}$ removes a deterministic *period- s season* ($\nabla_{(s)} s_t = s_t - s_{t-s} = 0$), *not* a polynomial trend. The two operators serve different purposes, and $\nabla_{(2)} \neq \nabla^2$.

Problem 2 (5 points)

(a) Expand the seasonal AR factor $(1 - \Phi B^4)X_t = (1 - \theta B)\varepsilon_t$ and solve for X_t :

$$\boxed{X_t = \Phi X_{t-4} + \varepsilon_t - \theta \varepsilon_{t-1}}$$

— an AR(1) acting only on the seasonal lag 4, driven by an MA(1) input. Invert the seasonal factor as a geometric series in B^4 (valid since $|\Phi| < 1$),

$$\frac{1}{1 - \Phi B^4} = \sum_{l \geq 0} \Phi^l B^{4l},$$

so that

$$X_t = \sum_{l \geq 0} \Phi^l B^{4l} (\varepsilon_t - \theta \varepsilon_{t-1}) = \sum_{l \geq 0} \Phi^l (\varepsilon_{t-4l} - \theta \varepsilon_{t-4l-1}).$$

Reading off the coefficient of ε_{t-j} :

$$\boxed{\psi_{4l} = \Phi^l, \quad \psi_{4l+1} = -\theta \Phi^l \quad (l \geq 0), \quad \psi_j = 0 \text{ otherwise.}}$$

(Check: $\psi_0 = 1$, $\psi_1 = -\theta$, $\psi_4 = \Phi$, $\psi_5 = -\theta\Phi$, $\dots$, and indeed $\psi_j - \Phi\psi_{j-4}$ reproduces the MA(1) numerator $1 - \theta B$.)

(b) Apply $\gamma_\tau = \sigma^2 \sum_{j \geq 0} \psi_j \psi_{j+\tau}$. The only nonzero taps sit at $j \equiv 0$ and $j \equiv 1 \pmod{4}$, so a product $\psi_j \psi_{j+\tau}$ survives only when both indices are of this type; this happens at $\tau \in \{0, \pm 1, \pm 3, \pm 4, \pm 5, \dots\}$, i.e. in clusters $\{4k-1, 4k, 4k+1\}$ around the seasonal multiples $4k$.

Lag 0: $\sum_l \psi_{4l}^2 + \sum_l \psi_{4l+1}^2 = \sum_l \Phi^{2l} + \theta^2 \sum_l \Phi^{2l} = (1 + \theta^2)/(1 - \Phi^2)$, so

$$\gamma_0 = \frac{(1 + \theta^2) \sigma^2}{1 - \Phi^2}.$$

Lag 1: the surviving pairs are $\psi_{4l} \psi_{4l+1} = \Phi^l (-\theta \Phi^l) = -\theta \Phi^{2l}$, so

$$\gamma_1 = \sigma^2 \sum_{l \geq 0} (-\theta \Phi^{2l}) = \frac{-\theta \sigma^2}{1 - \Phi^2}.$$

Lag 3: pairs $\psi_{4l+1} \psi_{4l+4} = (-\theta \Phi^l) (\Phi^{l+1}) = -\theta \Phi^{2l+1}$, so

$$\gamma_3 = \sigma^2 \sum_{l \geq 0} (-\theta \Phi^{2l+1}) = \frac{-\theta \Phi \sigma^2}{1 - \Phi^2}.$$

Lag 4: pairs $\psi_{4l} \psi_{4l+4} = \Phi^{2l+1}$ and $\psi_{4l+1} \psi_{4l+5} = (-\theta \Phi^l) (-\theta \Phi^{l+1}) = \theta^2 \Phi^{2l+1}$, so

$$\gamma_4 = \sigma^2 \sum_{l \geq 0} (1 + \theta^2) \Phi^{2l+1} = \frac{(1 + \theta^2) \Phi \sigma^2}{1 - \Phi^2} = \Phi \gamma_0.$$

Lag 5: pairs $\psi_{4l} \psi_{4l+5} = \Phi^l (-\theta \Phi^{l+1}) = -\theta \Phi^{2l+1}$, so

$$\gamma_5 = \frac{-\theta \Phi \sigma^2}{1 - \Phi^2} = \gamma_3.$$

Collecting,

$$\boxed{\gamma_0 = \frac{(1 + \theta^2) \sigma^2}{1 - \Phi^2}, \quad \gamma_1 = \frac{-\theta \sigma^2}{1 - \Phi^2}, \quad \gamma_3 = \gamma_5 = \frac{-\theta \Phi \sigma^2}{1 - \Phi^2}, \quad \gamma_4 = \Phi \gamma_0.}$$

So the seasonal AR(1) produces a spike at every seasonal lag $4k$ of size $\Phi^k \gamma_0$, each flanked by MA(1) side-lobes at $4k \pm 1$; the whole pattern decays geometrically in Φ .

(c) With $\Phi = \frac{1}{2}$, $\theta = \frac{2}{5}$, $\sigma^2 = 1$: $1 - \Phi^2 = \frac{3}{4}$ and $1 + \theta^2 = 1 + \frac{4}{25} = \frac{29}{25}$. Hence

$$\gamma_0 = \frac{29/25}{3/4} = \frac{116}{75} \approx 1.5467.$$

Then

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{-\theta}{1 + \theta^2} = \frac{-2/5}{29/25} = \frac{-2/5 \cdot 25}{29} = -\frac{10}{29} \approx -0.3448,$$

$$\rho_4 = \frac{\gamma_4}{\gamma_0} = \Phi = \frac{1}{2} = 0.5, \quad \rho_3 = \frac{\gamma_3}{\gamma_0} = \Phi \rho_1 = \frac{1}{2} \cdot \left(-\frac{10}{29}\right) = -\frac{5}{29} \approx -0.1724.$$

(So the correlogram shows a dominant seasonal spike $\rho_4 = 0.5$ with small negative side-lobes $\rho_3 = \rho_5 \approx -0.172$ at 4 ± 1 , and $\rho_1 \approx -0.345$ near the origin.)

Problem 3 (5 points)

(a) Since $X_t = \sum_{j=1}^t \varepsilon_j$ is a sum of mean-zero noise, $\mathbb{E}[X_t] = 0$. As the ε_j are uncorrelated with variance σ^2 ,

$$\text{Var}(X_t) = \sum_{j=1}^t \text{Var}(\varepsilon_j) = t \sigma^2.$$

For $\tau \geq 0$, $X_{t+\tau} = X_t + \sum_{j=t+1}^{t+\tau} \varepsilon_j$ adds only *new* (uncorrelated) noise, so the overlap is the first t shared terms:

$$\text{Cov}(X_t, X_{t+\tau}) = \text{Cov}\left(\sum_{j=1}^t \varepsilon_j, \sum_{k=1}^{t+\tau} \varepsilon_k\right) = \sum_{j=1}^t \sigma^2 = t \sigma^2.$$

Both $\text{Var}(X_t)$ and $\text{Cov}(X_t, X_{t+\tau})$ depend on t , so $\{X_t\}$ is **not** (weakly) stationary. However

$$\nabla X_t = X_t - X_{t-1} = \varepsilon_t,$$

which is white noise; hence $\{X_t\}$ is an ARIMA(0, 1, 0). Finally

$$\text{Corr}(X_t, X_{t+\tau}) = \frac{t\sigma^2}{\sqrt{t\sigma^2}\sqrt{(t+\tau)\sigma^2}} = \sqrt{\frac{t}{t+\tau}} \xrightarrow[t \rightarrow \infty]{} 1 \quad (\tau \text{ fixed}),$$

the near-unit correlation of nearby values characteristic of an integrated series.

(b) The recursion $Y_t = Y_{t-1} + (\varepsilon_t - \theta \varepsilon_{t-1})$ with $Y_0 = 0$ telescopes:

$$Y_t = \sum_{k=1}^t (\varepsilon_k - \theta \varepsilon_{k-1}) = \sum_{k=1}^t \varepsilon_k - \theta \sum_{k=1}^t \varepsilon_{k-1} = \sum_{k=1}^t \varepsilon_k - \theta \sum_{j=0}^{t-1} \varepsilon_j.$$

Separating the unmatched endpoints ε_t (only in the first sum) and ε_0 (only in the second),

$$Y_t = \varepsilon_t + \sum_{k=1}^{t-1} \varepsilon_k - \theta \varepsilon_0 - \theta \sum_{j=1}^{t-1} \varepsilon_j = \varepsilon_t + (1 - \theta) \sum_{j=1}^{t-1} \varepsilon_j - \theta \varepsilon_0,$$

as claimed. The coefficients of the $t + 1$ independent innovations $\varepsilon_0, \varepsilon_1, \dots, \varepsilon_t$ are: $-\theta$ on ε_0 , $(1 - \theta)$ on each of $\varepsilon_1, \dots, \varepsilon_{t-1}$ (that is $t - 1$ terms), and 1 on ε_t . By independence,

$$\boxed{\text{Var}(Y_t) = \sigma^2 \left[1 + (1 - \theta)^2(t - 1) + \theta^2 \right].}$$

For $\theta \neq 1$ this is *linear in t* with positive slope $(1 - \theta)^2 \sigma^2 > 0$: $\text{Var}(Y_t) = O(t)$, exactly the random-walk-like variance growth that flags an integrated (needs-differencing) series.

(c) At the boundary $\theta = 1$ the slope $(1 - \theta)^2$ vanishes, so

$$\text{Var}(Y_t) = \sigma^2 [1 + 0 + 1] = 2\sigma^2,$$

which is *bounded* (no growth). Indeed the recursion telescopes completely:

$$Y_t = \sum_{k=1}^t (\varepsilon_k - \varepsilon_{k-1}) = \varepsilon_t - \varepsilon_0,$$

so the level is itself stationary, and

$$\nabla Y_t = Y_t - Y_{t-1} = \varepsilon_t - \varepsilon_{t-1},$$

an MA(1) with unit coefficient $\theta = 1$. Its MA polynomial $1 - z$ has root $z = 1$ on the unit circle, so it is **non-invertible**, and its lag-1 autocorrelation is

$$\rho_1 = \frac{-\theta}{1 + \theta^2} = \frac{-1}{1 + 1} = -\frac{1}{2}.$$

This is the warning signature of **over-differencing**: differencing a series that was already stationary injects a $(1 - B)$ factor into the MA part, i.e. a non-invertible MA(1) with $\rho_1 = -\frac{1}{2}$. Seeing a sample $\rho_1 \approx -\frac{1}{2}$ after a difference is the cue to difference one time fewer.

Problem 4 (5 points)

(a) Multiply the model $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$ by X_{t-k} and take expectations. The process is causal, so X_{t-k} is a one-sided combination of $\varepsilon_{t-k}, \varepsilon_{t-k-1}, \dots$; hence for $k \geq 1$ the noise term is uncorrelated with the past, $\mathbb{E}[\varepsilon_t X_{t-k}] = 0$. With mean zero, $\mathbb{E}[X_{t-j} X_{t-k}] = \gamma_{k-j}$, giving for $k \geq 1$

$$\gamma_k = \phi_1 \gamma_{k-1} + \phi_2 \gamma_{k-2} \quad (\text{using } \gamma_{-1} = \gamma_1).$$

Multiplying instead by X_t keeps the noise term: by causality and $\psi_0 = 1$,

$$\mathbb{E}[\varepsilon_t X_t] = \mathbb{E}[\varepsilon_t (\phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t)] = \sigma^2,$$

so the variance identity is

$$\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma^2.$$

The single fact “causality $\Rightarrow \mathbb{E}[\varepsilon_t X_{t-k}] = 0$ for $k \geq 1$ ” is what closes the system. Replacing the population γ by the sample $\hat{\gamma}$ gives the Yule–Walker *estimators*.

(b) Taking $k = 1, 2$ gives the 2×2 Yule–Walker system $\hat{\Gamma}_2 \hat{\phi} = \hat{\gamma}_2$:

$$\begin{pmatrix} \hat{\gamma}_0 & \hat{\gamma}_1 \\ \hat{\gamma}_1 & \hat{\gamma}_0 \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \begin{pmatrix} \hat{\gamma}_1 \\ \hat{\gamma}_2 \end{pmatrix} \implies \begin{pmatrix} 10 & 5 \\ 5 & 10 \end{pmatrix} \begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}.$$

The inverse of $\begin{pmatrix} a & b \\ b & a \end{pmatrix}$ is $\frac{1}{a^2 - b^2} \begin{pmatrix} a & -b \\ -b & a \end{pmatrix}$; here $a = 10$, $b = 5$ and $a^2 - b^2 = 100 - 25 = 75$. Hence

$$\begin{pmatrix} \hat{\phi}_1 \\ \hat{\phi}_2 \end{pmatrix} = \frac{1}{75} \begin{pmatrix} 10 & -5 \\ -5 & 10 \end{pmatrix} \begin{pmatrix} 5 \\ 1 \end{pmatrix} = \frac{1}{75} \begin{pmatrix} 10 \cdot 5 - 5 \cdot 1 \\ -5 \cdot 5 + 10 \cdot 1 \end{pmatrix} = \frac{1}{75} \begin{pmatrix} 45 \\ -15 \end{pmatrix} = \begin{pmatrix} 0.6 \\ -0.2 \end{pmatrix}.$$

The innovation variance is

$$\hat{\sigma}^2 = \hat{\gamma}_0 - \hat{\phi}_1 \hat{\gamma}_1 - \hat{\phi}_2 \hat{\gamma}_2 = 10 - 0.6 \cdot 5 - (-0.2) \cdot 1 = 10 - 3 + 0.2 = 7.2.$$

So $\boxed{\hat{\phi}_1 = 0.6, \hat{\phi}_2 = -0.2, \hat{\sigma}^2 = 7.2}$ (Consistency check via $\gamma_0 = \sigma^2 / (1 - \phi_1 \rho_1 - \phi_2 \rho_2)$ with $\rho_1 = \frac{1}{2}, \rho_2 = \frac{1}{10}$: $1 - 0.6 \cdot \frac{1}{2} - (-0.2) \cdot \frac{1}{10} = 1 - 0.3 + 0.02 = 0.72$, and $7.2 / 0.72 = 10 = \hat{\gamma}_0$. ✓)

(c) The fitted AR polynomial is $\hat{\Phi}(z) = 1 - 0.6z + 0.2z^2$. Its roots solve $0.2z^2 - 0.6z + 1 = 0$, i.e. $z^2 - 3z + 5 = 0$:

$$z = \frac{3 \pm \sqrt{9 - 20}}{2} = \frac{3 \pm \sqrt{-11}}{2} = \frac{3}{2} \pm \frac{\sqrt{11}}{2} i, \quad |z| = \sqrt{\left(\frac{3}{2}\right)^2 + \frac{11}{4}} = \sqrt{\frac{9}{4} + \frac{11}{4}} = \sqrt{5} \approx 2.236.$$

Both roots have modulus $\sqrt{5} > 1$ (outside the unit circle), so the fit is **causal/stationary**. Equivalently, the AR(2) stationarity-triangle conditions hold: $\hat{\phi}_1 + \hat{\phi}_2 = 0.4 < 1$, $\hat{\phi}_2 - \hat{\phi}_1 = -0.8 < 1$, and $|\hat{\phi}_2| = 0.2 < 1$. (Unlike least squares, the Yule–Walker estimator always returns a stationary fit.)